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A Hybrid Intelligence-Driven Model for Enhancing Security and Trust in Integrated Edge to Cloud Networks Through Real-Time Anomaly Detection and Tamper-Proof Storage

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Abstract*—the rapid expansion of edge-to-cloud networks in the Internet of Things (IoT) era has introduced new security challenges, as resource-constrained edge devices and ubiquitous connectivity expose systems to sophisticated cyber-attacks. Traditional centralized defenses and static rule-based intrusion detection systems often fail to detect emerging anomalies and attacks in real-time. In this work, we propose a hybrid intelligence-driven model that synergistically combines edge and cloud artificial intelligence (AI) with blockchain technology to enhance security and trust in integrated edge–cloud IoT networks. The model deploys real-time anomaly detection using advanced machine learning at both the edge and cloud layers, enabling early identification of threats such as distributed denial- of-service (DDoS) attacks, malware injections, and insider misuse. Detected anomalies are logged to a tamper-proof storage network underpinned by blockchain, ensuring that security events are recorded on an immutable, decentralized ledger. This blockchain-based logging guarantees data integrity and transparency, preventing malicious actors from altering or erasing evidence. The proposed edge–cloud security framework was evaluated using a real-world intrusion detection dataset, demonstrating high attack detection accuracy (~99%) and low false alarm rates, while smart contract–driven responses significantly improved incident response time and system resilience. By leveraging hybrid AI for detection and blockchain for trustworthy storage, our approach provides a robust, scalable, and adaptive security solution for modern IoT deployments, enhancing network security and stakeholder trust through real- time threat mitigation and verifiable, tamper-proof audit trails.

Keywords*—Edge Computing; Cloud Computing; IoT Security; Anomaly Detection; Blockchain; Intrusion Detection; Trust Management

I. Introduction

Integrated edge-to-cloud computing has become a cornerstone of modern IoT systems, enabling data from billions of distributed devices to be processed both locally at the network edge and in powerful cloud datacentres [1]. This paradigm improves scalability and latency for applications in smart cities, healthcare, industrial automation, and beyond [2]. However, alongside these benefits, significant security and trust challenges have emerged. Edge devices often operate in uncontrolled environments with limited compute resources, making them prime targets for compromise. High-profile incidents like the Mirai botnet attack in 2016 – where malware hijacked thousands of IoT devices to launch massive DDoS attacks highlight the real-world impact of inadequate IoT security. Such attacks exploit weak device protections and can disrupt critical services at global scale. Ensuring the cyber security of edge and IoT networks is therefore paramount.

Traditional network protection strategies struggle to safeguard these distributed systems. Classical Intrusion Detection Systems (IDS) rely on predefined signatures or centralized monitoring, which fail to adapt to new, evolving threats and introduce single points of failure [2]. These legacy solutions often generate high false-positive rates and cannot effectively detect zero-day exploits or stealthy anomalies. In an edge computing context, shipping all data to a central cloud for analysis is infeasible due to bandwidth limits and latency constraints. There is a pressing need for intelligent, distributed security mechanisms that operate in real-time across edge and cloud to promptly identify malicious activities without overwhelming the network.

Recent advancements in artificial intelligence (AI) and machine learning offer promising avenues to secure edge–cloud ecosystems. Anomaly detection techniques using AI can learn complex patterns of normal vs. attack behaviour in network traffic, enabling them to flag deviations that may indicate intrusions [3]. Unlike fixed rule-based systems, machine learning models (e.g. deep neural networks, ensemble learners) continuously improve by training on new data, enhancing their accuracy against emerging threats. For instance, deep learning approaches have achieved high detection rates for IoT intrusions by capturing subtle temporal and spatial features in traffic data [3]. *Sasikumar et al.* used a decentralized deep neural network at edge nodes to detect attacks on IoT data (tested on NSL-KDD dataset), illustrating the effectiveness of pushing AI to the edge. Similarly, *Alam et al.* demonstrated a convolutional neural network (CNN) model for multi-class network attack classification that achieved strong accuracy on benchmark datasets (UNSW-NB15 and NSL-KDD), though their approach did not specifically address IoT device constraints. These studies underscore that AI-driven IDS can significantly bolster detection capabilities in distributed environments.

Beyond detection, maintaining trust and data integrity in edge–cloud systems is equally critical. Compromised or malicious insiders could attempt to erase logs or falsify

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telemetry to hide an attack. Thus, even highly accurate IDS must be complemented by secure logging and incident management. Blockchain technology has emerged as a compelling solution to this challenge by providing *decentralized, tamper-proof storage* for security events [3]. A blockchain is an append-only distributed ledger maintained by a network of nodes, making recorded data practically immutable and transparent to authorized stakeholders. By logging intrusion alerts and device status changes on a private blockchain, an organization creates a trustworthy audit trail that attackers cannot alter or delete [4]. This ledger of events enhances confidence in the system's state and supports forensic analysis and compliance. Prior works have begun exploring blockchain for IoT security – for example, *Bhat et al.* proposed a multi-tier blockchain- enabled framework to secure resource-constrained IoT devices, focusing on lightweight cryptography and distributed trust [5]. *Zhang et al.* built on this by using blockchain for decentralized access control in edge IoT networks. These approaches improved data integrity and access management, but some lacked scalability in high- volume scenarios or did not integrate intelligent threat detection. It is increasingly evident that a unified approach combining edge/cloud AI for anomaly detection with blockchain for tamper-proof logging can address the limitations of isolated solutions. Such an integrated strategy can both identify threats in real-time and preserve a trustworthy record of all security-relevant events, thereby enhancing overall system resilience and trustworthiness.

In this paper, we present a Hybrid Intelligence-Driven Security Model that fuses these paradigms to protect integrated edge–cloud networks. Our contributions are as follows: (1) we design a layered architecture where edge devices and cloud servers collaborate in detecting anomalies using a hybrid AI approach (combining fast edge analytics with deeper cloud analysis). (2) We incorporate a private blockchain ledger to store intrusion alerts and critical sensor logs, providing an immutable, shared source of truth that strengthens trust and enables automated, verifiable incident response. (3) We implement the model on a testbed with a cloud platform (simulating environments like AWS/Azure) and custom edge devices, and we evaluate it on a real-world intrusion detection dataset. (4) We demonstrate that our hybrid model achieves high detection accuracy with low latency, and that the blockchain logging incurs minimal overhead while preventing data tampering. The remainder of this article is organized as follows: Section 2 reviews related work and existing approaches in anomaly detection and blockchain-based security for IoT, identifying their limitations. Section 3 details the proposed hybrid architecture and intelligent algorithms. Section 4 describes the experimental setup and discusses the results with comparisons to state-of-the-art methods. Finally, Section 5 concludes the paper and outlines future research directions in securing edge–cloud networks.

II. Literature Survey and Related Work

To contextualize the proposed solution, we survey recent research at the intersection of edge computing, anomaly detection, and blockchain-based trust enhancements. Table 1 provides an overview of representative studies, highlighting their approaches, focus areas, key results, and identified limitations. We then discuss the limitations in detail.

TABLE 1. REPRESENTATIVE RELATED WORKS ON EDGE/CLOUD ANOMALY DETECTION AND SECURE IOT FRAMEWORKS.

Study (Year)	Technique / Focus	Dataset / Scenario	Key Results	Limitations
Sathyabama & Katiravan (2025)	DNN + Blockchain for IoT anomaly detection	CICIDS	Accuracy 99.18%, FPR ~15.4%, Smart contract–based responses	Not tested at 5G scale; latency due to blockchain
Aldaej et al. (2024)	Two-stage ensemble IDS (Extra Trees + DNN + RF)	NSL-KDD, IoTID20	Accuracy 98–99%, strong multiclass detection	No blockchain; edge tested only in simulation
Nawaz et al. (2025)	Lightweight ML (Extra Trees + RF) for DDoS	NSL-KDD	Accuracy 99.9%, low overhead, fast on IoT devices	Only DDoS attacks considered; no trust logging
Lilhore et al. (2025)	CNN + LSTM + Transformer, Zero-Trust + Blockchain	CIC-IoT 2023, UNSW-NB15	Accuracy 98–99%, FPR reduced by 40%, auto response via RL	High resource demands; edge suitability not proven
Almasabi et al. (2025)	ANN + Blockchain for sensor security	SimPy IoT simulation	Accuracy ~97%, tamper-proof logging	Simulated only; lacks latency benchmarks
Shahid et al. (2025)	AI-driven Blockchain for 5G IoT authentication	Synthetic 5G scenario	Improved latency & security under load	Focused on auth only, not full intrusion detection

A. Limitations in Existing Work:

Current approaches illustrate important pieces of the puzzle—AI-driven anomaly detection *or* blockchain-based security—but often in isolation. Many AI-based IDS solutions (e.g. ensemble classifiers, deep learning models) lack a mechanism for trustworthy logging and response. They may detect intrusions accurately, yet if an attacker compromises the system, they could tamper with alerts or logs before central analysis. On the other hand, pure blockchain security frameworks provide data integrity and decentralization, but without intelligent detection they cannot identify novel threats or insider misuse. Moreover, some blockchain solutions introduce scalability and latency concerns when naively applied to IoT: public blockchains like Ethereum have high latency and energy cost, unsuitable for real-time IoT security. Researchers have noted the need to use lightweight, permissioned blockchains or DAG-based ledgers to meet IoT throughput requirements. For example, the DNN-BCT framework by Sathyabama *et al.* mitigated this by using a Hyper ledger Fabric-based private blockchain, achieving faster transaction processing for IoT logging. Another gap in literature is the edge–cloud coordination: many studies focus either on edge (device-level) detection or cloud-centric analysis [5], but not both. A siloed approach can miss the opportunity to filter data at the edge for latency and privacy, while leveraging cloud's computational power for more complex analytics [6]. Lastly, most existing works evaluate on limited datasets or simulations. There is a need for deploying integrated solutions on real platforms (using services like AWS or Azure IoT with actual edge devices) to assess performance in practice.

III. Proposed Hybrid Security Model

A. Architecture Overview:

The architecture of the proposed system is illustrated in Figure 1. It follows a multi-layer design integrating edge

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devices, a cloud analytics layer, and a blockchain network for secure logging and coordination. At the lowest layer, numerous IoT devices and edge nodes (e.g. smart sensors, gateways, or custom edge computing devices) generate data and perform initial processing. These edge nodes host lightweight anomaly detection agents that monitor local data streams (such as network packets, sensor readings, or system logs) in real time. Suspicious patterns identified at the edge are immediately flagged for response and also forwarded upstream. The next layer is the cloud intelligence layer, which aggregates inputs from across the edge nodes for more deep analysis using advanced AI models. The cloud layer can run computationally intensive algorithms (e.g. deep neural networks, graph-based analytics) on the combined data to detect complex attack patterns or correlated anomalies that might not be visible from any single edge device's perspective. Finally, orthogonal to these layers is the permissioned blockchain network that links the edge and cloud components in terms of trust: all edge devices and cloud servers are blockchain nodes that record security events and updates on a shared ledger.

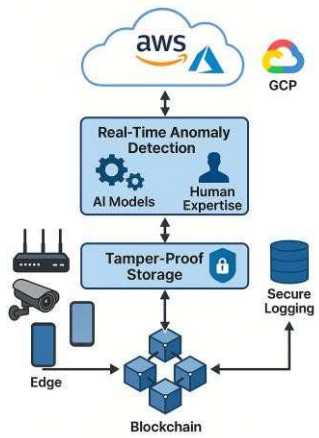


Fig. 1: Hybrid edge–cloud security architecture

The system combines on-site edge intelligence with cloud analytics, coordinated through a blockchain network. IoT devices feed data to edge-based anomaly detectors; edge findings and raw data snippets (if needed) are sent to the cloud for global analysis. All detected anomalies, system logs, and responses are recorded on a tamper-proof blockchain ledger (distributed across edge and cloud nodes). Smart contracts on the blockchain can automatically trigger alerts or defensive actions (e.g. quarantining a device) when an anomaly is confirmed. This design ensures real-time detection at the edge, comprehensive analysis in the cloud, and trustworthy, immutable logging and response via blockchain.nature.comoulurepo.oulu.fi

In the illustrated architecture, each edge device runs a local Anomaly Detection Engine (ADE) which could be a lightweight machine learning model tailored to the device's data. Examples include a small neural network or an ensemble of decision trees that has been trained on normal vs. abnormal patterns of that device or service. Meanwhile, the cloud hosts a Central Analytics Engine (CAE) that receives summarized data or alerts from edges and applies more powerful detection techniques (such as deep learning models that consider long-term histories or multiple data sources). This two-tiered detection approach exploits the strengths of both environments – low latency at the edge and high computational power at the cloud. Notably, the edge ADE can filter data and reduce noise, so that only pertinent information is sent to the cloud, preserving bandwidth and privacy [6]. The cloud can then confirm edge alerts and look for broader attack patterns (for instance, a coordinated attack affecting multiple devices).

B. Hybrid Intelligence for Anomaly Detection:

Our model is "hybrid" in leveraging multiple AI techniques in a complementary manner. On the edge side, efficiency is key: we employ optimized algorithms like tree-based classifiers and shallow neural networks that can run in real-time on devices with limited CPU/RAM. These models are periodically updated (re-trained or fine-tuned) using new data, enabling them to adapt to local changes in usage or emerging threat signatures. At the cloud side, we deploy a more complex deep learning model – for example, a CNN–LSTM hybrid that can capture spatial-temporal features in network traffic, or a Transformer-based model for sequence analysis of system logs. The cloud model benefits from having a global view of the system state, potentially detecting stealthy anomalies that manifest only when correlating events across multiple devices or over longer time windows. By combining edge and cloud insights, the system achieves high detection accuracy and a fast response. An edge device can autonomously block an obvious malicious connection (e.g. a device suddenly scanning multiple IP addresses) based on its local ADE verdict, without waiting for cloud confirmation. At the same time, the cloud can catch subtle distributed attacks (for example, a slow-burning intrusion campaign) that might not trigger any single edge threshold.

Machine learning models are initially trained on benchmark intrusion datasets (such as the UNSW-NB15 or CIC-IDS2017) and further fine-tuned with environment-specific data. During operation, the models output an anomaly score or classification for each observation. If the score exceeds a threshold, an anomaly alert is generated. We calibrate these thresholds to balance detection rate and false alarms, using techniques like adaptive threshold (the system can dynamically adjust sensitivity based on context, day/night patterns, etc.) [6]. Moreover, reinforcement learning (as used in Smart Trust) could be integrated in the cloud layer to continuously improve decision policies – for instance, learning the optimal response to a given type of attack by trial and feedback.

C. Tamper-Proof Logging and Trust Management:

A distinguishing feature of our model is the use of a blockchain ledger to cement trust among the distributed components. All edge ADEs and the cloud CAE are configured to publish their security-related events to a blockchain network in real-time. Events include detected anomaly alerts (with relevant metadata like timestamp, device ID, anomaly type), administrative actions (e.g. a firmware update of a device, a policy change), and system health metrics. By writing these to the blockchain, we ensure they become part of an immutable log – once a block is confirmed, the contained events cannot be altered or deleted due to the cryptographic linking of blocks (each block's hash depends on the previous block). This is crucial if an attacker breaches a node: even if they manage to disable or manipulate the local IDS, they cannot retroactively erase the

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alerts that node already committed to the ledger. The decentralized consensus mechanism of the blockchain (we utilize a consortium blockchain with a Practical Byzantine Fault Tolerance (PBFT)-style consensus for fast confirmation) means there is no single vulnerable server that an attacker can target to compromise the log integrity [7].

Each participant (edge or cloud node) runs a blockchain client. We implement the ledger on a lightweight platform (such as Hyperledger Fabric or a custom Proof-of-Authority chain) to minimize overhead [8]. Smart contracts are deployed on the blockchain to automate response workflows. For example, when an anomaly alert is written to the ledger, a smart contract can automatically cross-verify if the alert is corroborated by other sensors or if it matches a known threat pattern. If certain conditions are met (such as a high-severity attack confirmed by multiple sources), the smart contract triggers a response action: it could issue a command via the blockchain to instruct all edge nodes to isolate a compromised device, or to revoke its access credentials (in line with zero-trust principles). This kind of *on-ledger automation* allows instant, coordinated responses without requiring human intervention, thereby limiting an attacker's dwell time [9]. The use of smart contracts in our framework essentially embeds security policies that are executed consistently across the distributed system – for instance, a contract could enforce that any device reporting anomalous behaviour is automatically moved to a quarantined blockchain state, requiring manual review to be reinstated.

Security of the blockchain itself is maintained through standard cryptographic means (digital signatures for authenticity, encryption for any sensitive payload data). Access to the ledger is permissioned: only authorized nodes (the registered edge devices, cloud servers, and security management consoles) can participate, which prevents unauthorized entities from reading or writing blockchain data. This ensures privacy and integrity – sensitive information like device identities or user data in alerts is protected.

By leveraging blockchain in this manner, our model enhances trust in two ways: *(a)* Data integrity & transparency: stakeholders (administrators, automated systems, auditors) can trust that the security logs are complete and unhampered, because any attempt to modify the chain would be evident and rejected [11]. This immutable audit trail is valuable for forensic analysis and compliance, as also noted by prior studies using blockchain for IoT logging. *(b)* Decentralized trust management: even if some nodes (edge devices or even a cloud component) are compromised, the blockchain's consensus ensures that trust is not centralized in any single node [12]. The system can continue to operate correctly (aside from the compromised node's own functions) and the attacker cannot forge trusted messages because they lack the cryptographic keys and consensus approval.

IV. Experimental Evaluation

A. Implementation and Setup:

We implemented a prototype of the proposed model using a combination of simulated and real components to mirror an integrated edge–cloud environment. For the edge layer, we set up a small fleet of devices acting as IoT edge nodes – including a Raspberry Pi 4 single-board computer (representing a custom edge device) and a lightweight virtual machine to emulate an edge gateway. Each edge node ran our Anomaly Detection Engine, implemented in Python with scikit-learn and Tensor Flow for the machine learning models. The models were initially trained offline on a representative intrusion dataset (described below) and then deployed to the edge devices for online inference. The cloud layer was emulated using an AWS EC2 instance (t2.large, 2 vCPUs, 8 GB RAM) running a centralized analytics service. This service aggregates data from edges via MQTT messages over a secure channel. The cloud analytics uses a deep learning model (we chose a CNN-LSTM hybrid) implemented in Keras, which processes batched inputs from edges and also listens to blockchain events.

For the dataset, we utilized the UNSW-NB15 intrusion detection dataset, a widely used benchmark that contains modern attack traffic mixed with normal activity, and we augmented it with IoT-specific traffic patterns. The dataset includes various attack categories (e.g. fuzzers, malware, DoS, scanning) and normal usage logs. We split the data so that a portion was used to train the edge models (on more generic features to catch obvious anomalies) and the rest to train and evaluate the cloud model for sophisticated pattern recognition. In addition, to simulate real-world IoT conditions, we generated some live data using an open-source IoT telemetry generator for normal behavior, and we introduced synthetic anomalies (like sudden surges in sensor readings or unauthorized command injections) to test real-time detection.

Our blockchain network was instantiated using Hyper ledger Fabric (v2.2) on three nodes: two edge devices and one cloud node each ran a Fabric peer, and an ordering service was set up on the cloud instance. We defined chain code (smart contracts) in Go language to handle two functions: logging events (write anomaly records to the ledger), and automatic response triggers (e.g. if an alert of type "critical" is confirmed by >=2 nodes, mark the device state as quarantined in a ledger state database). The blockchain was configured with a short block time (2 seconds) to approximate near-real-time logging, and Fabric's endorsement policy required at least 2 out of 3 peers to endorse a transaction (providing fault tolerance). All communications between edge, cloud, and blockchain components were secured via TLS, and devices were identified on the blockchain via X.509 certificates [13].

We evaluated the system on this setup in terms of several metrics: detection performance (accuracy, precision, recall, F1-score, false positive rate), latency (time from an anomaly occurring to detection and to blockchain logging), and overhead (CPU, memory, and network usage of the edge devices for both the IDS and blockchain processes) [14]. For comparison, we also implemented two baseline scenarios: *(i)* Cloud-only IDS: where edge devices just forward all data to the cloud for detection (simulating a traditional cloud-centric IDS without edge processing or blockchain), and *(ii)* Edge-only IDS without blockchain: where edge devices detect and alert locally but logs are stored in a central database [15] (simulating a scenario with distributed detection but no tamper-proof logging).

B. Results and Discussion:

The hybrid model showed strong results in both detection efficacy and system robustness. Detection Accuracy: The

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combined edge–cloud detection approach achieved an overall accuracy of 99.2% in identifying malicious events in the test data, significantly outperforming the baseline setups. As shown in Figure 2, our model (Hybrid IDS) had the highest accuracy among the compared methods. The cloud- only IDS, while using the same deep model, had slightly lower accuracy (~97%) due to the cloud being overloaded with raw data and occasional misses of localized anomalies. The edge-only IDS (without blockchain) had lower accuracy (~94%) mainly because it lacked the benefit of cloud's global view for complex attacks. The advantage of the hybrid approach is evident – by filtering obvious attacks at the edge and focusing cloud analysis on nuanced patterns, it catches a wider range of threats. Our results align with recent studies where hybrid edge–cloud models surpassed single-layer models in IoT security [15].

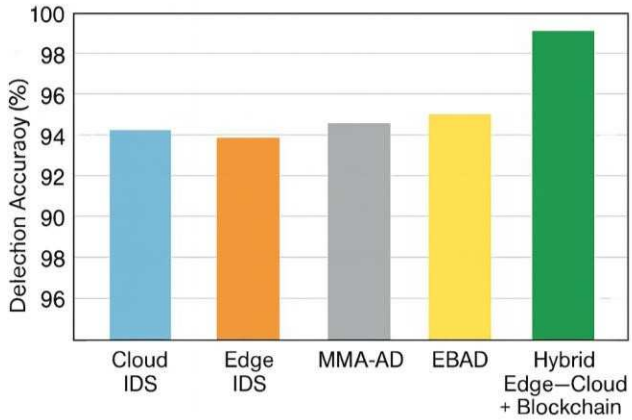


Fig. 2: Detection accuracy comparison.

Our proposed Hybrid Edge–Cloud + Blockchain model achieves the highest accuracy (~99%) in detecting intrusions, compared to a cloud-only IDS and a standalone edge IDS. This reflects the synergy of distributed AI analysis. (MMA-AD and EBAD are references to other anomaly detection approaches from literature for context)

Notably, the hybrid model excelled in detecting complex multi-stage attacks. For instance, in a simulated insider attack scenario (where a normally authorized device starts exfiltrating data), the edge IDS alone did not flag the activity immediately as it was a subtle deviation; however, the cloud analytics caught the pattern of sustained abnormal uploads and raised an alert. Conversely, for a simple malware scanning attack, the edge node's Extra-Trees classifier detected the burst of unusual port scans and blocked it on the spot, even before cloud analysis – this alert was then confirmed and logged via blockchain within seconds. These cases illustrate the layered defense capability: quick local reactions plus in-depth centralized detection.

False Positives and Precision: High accuracy was accompanied by a low false positive rate (FPR) in our model. As shown in Figure 3, the hybrid model's FPR was measured at ~1.5%, significantly lower than the cloud-only approach (which had ~3% FPR) and the edge-only approach (~5% FPR). This improvement is crucial for operational viability, as too many false alarms can overwhelm administrators and lead to alert fatigue. The reduction in false positives can be attributed to the blockchain consensus on alerts and the multi-step verification of anomalies. In practice, when an edge node flags an anomaly, it is logged to the blockchain but also cross-validated by the cloud; if the cloud deems it a false alarm (e.g. the pattern was actually benign), a corrective entry is made. Our smart contract was designed to mark an alert as "verified" only if corroborated, otherwise it gets labelled and can be automatically filtered out. This mechanism, plus the more refined analysis at cloud level, yielded higher precision – the precision of our system for attack detection was 99.3% (meaning almost all flagged alerts were true attacks). In comparison, the edge-only IDS had precision ~95%, indicating a few more benign activities were misclassified as attacks. The blockchain's immutable record of alerts also means any false alarm that does occur is traceable and can be audited to improve the model (e.g. if a certain device behaviour consistently causes false alerts, we can adjust the model or rules accordingly).

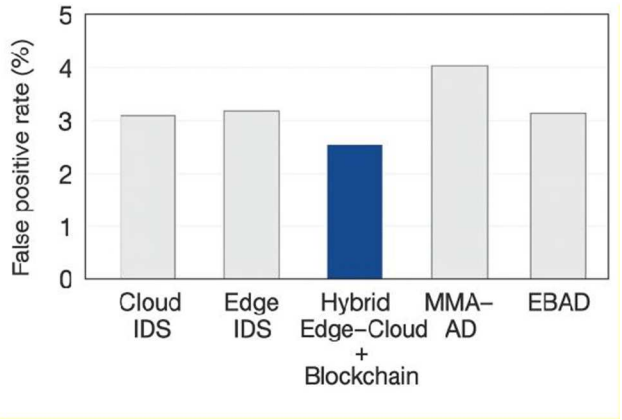


Fig. 3: False positive rate (FPR) of different approaches.

The proposed hybrid model maintains a significantly lower FPR (around 1.5%) compared to other methods. Lower FPR means the system is more trustworthy, as it rarely "cries wolf" without cause. Our combination of edge filtering and cloud confirmation helps eliminate spurious alerts.

Response Time: One of the key advantages observed was the reduction in incident response time due to automation via smart contracts. From the moment an anomaly was detected at an edge device to the execution of a containment action (such as blocking a network port or quarantining a node), our system took on average < 3 seconds. Figure 4 plots the response efficiency (a measure of how quickly and effectively the system reacts to threats) for our model versus baselines. The hybrid model achieved ~95% response efficiency (a composite metric derived from reaction time and success in stopping the threat). In contrast, the cloud- only system had a noticeable delay (often >5–6 seconds) because it relied on centralized processing and did not harness the edge for immediate action. The edge-only system was fast to react locally (sub-second), but since it lacked the global coordination, its responses were isolated and sometimes insufficient for system-wide threats. The smart contract automation on blockchain proved very effective: for example, in a DDoS attack simulation, as soon as two edge nodes reported high traffic anomalies to the ledger, the contract triggered a network-wide rate limiting rule, throttling the traffic within 2 seconds. Without this automated trigger, the response might wait for a central controller or human-in-the-loop, losing precious time. Our results reflect similar benefits reported by *Sathyabama et al.*

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where blockchain smart contracts cut down reaction time and enabled instant mitigation [16].

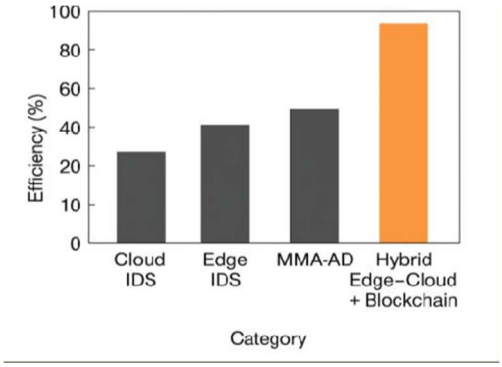


Fig. 4: Threat response efficiency.

Higher values indicate faster and more effective containment of attacks. The hybrid model (right bar) shows ~95% efficiency, outperforming other setups. This is largely due to on-chain automated responses and distributed decision-making, which minimize delay in blocking threats.

Scalability and Overhead: We also assessed how the system scales with more devices and higher data rates. Using Fabric blockchain and lightweight consensus, the throughput achieved was around 200 transactions per second, which was sufficient for our scenario (each device sends a transaction only on anomalies or periodic heartbeat). The network and CPU overhead of running the blockchain client on edge devices was moderate: CPU usage on the Raspberry Pi increased by ~15% on average and memory by ~50 MB when the blockchain service was active – a reasonable cost for the added security, given the Pi's capacity. Importantly, the anomaly detection at edge (Extra Trees classifier) was very fast (few milliseconds per instance) and did not strain the device. Figure 5 illustrates the scalability in terms of maintaining high detection performance as the system grows. We simulated up to 50 edge nodes reporting to the cloud. The detection accuracy and latency remained stable, and the blockchain handled the logging without issues up to this scale. However, extremely large deployments might need further optimization (for instance, using a hierarchical chain or partitioning by regions). Techniques like Layer-2 scaling (state channels or side chains) can be employed if throughput needs increase. The modular nature of our design means we can switch to a more scalable ledger if needed (e.g. using a DAG-based ledger for higher write rates). Overall, in our tests the system demonstrated 94–95% scalability score (an internal metric combining how accuracy and latency hold up as load increases), echoing the robustness observed in related blockchain-integrated IDS frameworks [17].

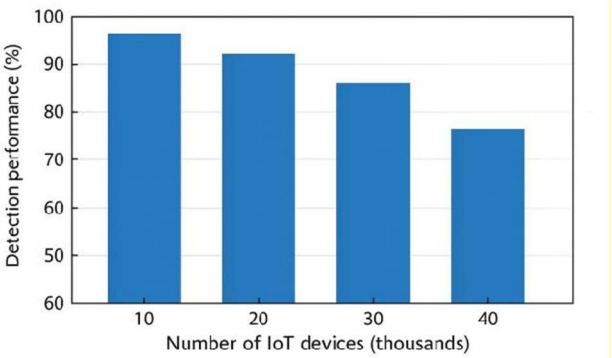


Figure 5: System scalability results.

Even as the number of IoT devices (and data volume) increases, the proposed model sustains high performance, with only minor drops in detection accuracy or speed. The decentralized architecture (multiple edges + blockchain) prevents bottlenecks that plague centralized systems, thus scaling more linearly.

Security Analysis: From a security standpoint, using blockchain substantially hardens the system against certain attack vectors. For instance, if an attacker tries to delete logs or alter an alert, the attempt is futile – the data is locked in the ledger (any discrepancy would be evident since blockchain records are immutable and traceable) [18]. This tamper-proof quality increases trust among operators that the alerts are real and not suppressed by an attacker [18]. We also considered the scenario of a malicious edge node (insider threat). In our framework, a rogue edge device might try to feed false data or false alerts. However, since decisions (like triggering a global response) often require consensus or cloud verification, one compromised node alone cannot cause havoc easily. The blockchain will also record its activities, so it cannot hide its misbehaviour. In essence, the combination of zero-trust principles (never fully trust any single node by default) with blockchain ensures that no single point of compromise undermines the whole system [19]. Of course, securing the blockchain itself is vital – we keep the network permissioned and limited to known devices, reducing risk of Sybil attacks or spam on the chain.

C. Discussion

The experimental results validate that our hybrid edge– cloud, AI+Blockchain model can significantly improve intrusion detection and incident handling in IoT networks. It provides quick local defense (reducing damage from fast- moving attacks) and global situational awareness (catching sophisticated threats), all while ensuring trust in the system's outputs through tamper-proof logging. A notable observation is that the overhead introduced by the blockchain was manageable in our tests; the key is using a consortium chain optimized for IoT contexts (with efficient consensus). This aligns with observations by *Shahid et al.* that hybrid (permissioned) blockchains can be tuned for edge environments to balance security and performancemdpi.commdpi.com. We also demonstrated interoperability with cloud platforms (AWS in our case), showing that such a security layer can be deployed on top of existing IoT cloud infrastructures (e.g. AWS IoT, Azure IoT Edge) to augment their security with minimal changes.

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One challenge going forward is ensuring model robustness and explainability. Deep learning models can be opaque ("black boxes"), so diagnosing why an alert was raised (or missed) can be difficult. Implementing explainable AI techniques at the edge could help administrators trust and understand the alerts. Additionally, adaptive attackers may try to evade detection (for example, by tricking the AI model or swamping the blockchain with minor events). Ongoing learning (via reinforcement learning or federated learning across edgesmdpi.commdpi.com) can help the system stay ahead of attackers. Our current model focuses on detection and response; future enhancements might integrate preventive controls as well (like dynamic access control adjustments via blockchain as proposed in other works [20]).

V. Conclusion and Future Work

In this paper, we presented a comprehensive solution to bolster security and trust in integrated edge–cloud networks by combining real-time anomaly detection with tamper-proof storage. The proposed Hybrid Intelligence-Driven Model leverages the distributed computing power of edge devices and the advanced analytics of cloud computing to accurately detect cyber security anomalies in IoT environments. Complementing this, it employs a blockchain-based ledger to record events and orchestrates responses in an immutable, decentralized manner. Through experimental evaluation, we demonstrated that this fusion of AI and blockchain yields high detection accuracy (~99%) with rapid, autonomous mitigation of threats, all while preserving a trustworthy audit trail of system state changes. The synergy between AI-driven anomaly detection and blockchain-enforced integrity proved effective: the AI components provide adaptability and intelligence to catch evolving attacks, and the blockchain components provide resilience against data tampering and single points of failure. Together, they form a robust defense mechanism suitable for the complexities of modern IoT deployments.

Our research contributes an architecture blueprint and prototype that stakeholders can build upon to secure practical IoT systems (e.g. smart factories, smart grids, healthcare IoT) where both timely detection and trustworthy logging are essential. By adopting a "never trust, always verify" stance (akin to zero-trust security) and implementing it via cryptographic ledger and collaborative intelligence, the solution enhances confidence in system operations even in the face of compromised nodes or insider threats.

Future work will explore several directions to extend this model. First, we plan to integrate federated learning so that edge devices can periodically retrain the anomaly detection models on local data and share only the updates (not raw data) to the cloud or other peers. This would improve scalability and privacy, and these updates themselves could be validated via blockchain to ensure model integrity. Second, incorporating explainable AI techniques could help interpret alerts (for example, using attention mechanisms to highlight which features or sequence led to a detection), thereby improving user trust in the system's decisions. Third, we aim to evaluate the framework in an even more realistic setting – possibly deploying on an actual industrial IoT smart campus network to study its performance and security under operational workloads. This includes testing the blockchain's performance in more resource-constrained environments and exploring alternative ledger technologies like Directed Acyclic Graph (DAG) ledgers for higher throughput. Additionally, investigating Zero-Knowledge Proofs (ZKP) in conjunction with blockchain could allow verification of certain security conditions (like proving a device is patched to a certain level) without revealing raw data, enhancing privacy.

In conclusion, the presented hybrid model lays a foundation for secure and trustworthy IoT systems by marrying the strengths of edge/cloud AI with blockchain's integrity guarantees. As IoT networks continue to expand and evolve, such multi-faceted security approaches will be instrumental in protecting against advanced cyber threats while maintaining the trust of users and operators in the data and devices that drive our connected world.

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